Energy Baseline Analytics Report

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manual_overrides: None recorded

skipped_families: None recorded

assumption_fallbacks: {'alpha': 0.05, 'assumptions_checked': [{'action': 'full_sample_retained', 'metric_name': 'original_n', 'metric_value': 36.0, 'name': 'large_n_subsampling_policy', 'passed': True, 'threshold': 4000}, {'action': 'proceed_adf_kpss', 'metric_name': 'std', 'metric_value': 3171.830633, 'name': 'non_degenerate_variance', 'passed': True, 'threshold': 1e-12}, {'action': 'switched_to_hac_newey_west', 'metric_name': 'lag1_r', 'metric_value': 0.4195, 'name': 'serial_independence', 'passed': False, 'threshold': 0.2}], 'assumptions_met': False, 'ci': {'confidence_level': 0.95, 'estimate': 0.426212, 'high': 0.69053419, 'inference_label': 'hac_robust', 'low': 0.16188982, 'method': 'hac_ar1_persistence_t(bw=3)', 'std_error': 0.1223505}, 'correction': 'holm', 'decision': 'fail', 'decision_rule': 'p_only', 'details': {'adf_lags': 10, 'adf_p': 0.39851449, 'adf_statistic': -1.76392248, 'autocorrelation_check': {'autocorrelated': True, 'lag1_r': 0.4195, 'ljung_box_p': 0.0002612, 'n': 36}, 'kpss_lags': 1, 'kpss_p': 0.1, 'kpss_statistic': 0.11648566, 'lock_custom_matrix_decision': True, 'matrix_cell': 'inconclusive_low_power', 'matrix_force_fail': True, 'policy_evaluation_note': 'p=0.025 <= alpha=0.05 (p_only).', 'subsampling': {'applied': False, 'max_fast_n': 4000, 'original_n': 36, 'strategy': 'full_sample_exact', 'used_n': 36}}, 'df': 13.0, 'downstream_action': 'use_robust_differencing_sensitivity', 'effect_size': 0.426212, 'effect_size_name': 'ar1_persistence_abs_phi1', 'fallback_used': 'hac_newey_west', 'family': 'stationarity_trend', 'inference_label': 'hac_robust', 'interpretation': 'Neither ADF nor KPSS rejects -> low-power / short-span inconclusive stationarity. [artifact:annotation.assumption_fallbacks.0]; (ADF=-1.764, p_ADF=0.3985; KPSS=0.116, p_KPSS=0.1; |phi_1|=0.426).', 'library_versions': {'energybaseline': '3.0.0', 'numpy': '2.4.6', 'scipy': '1.17.1', 'statsmodels': '0.15.0', 'sklearn': '1.9.1', 'pandas': '3.1.1'}, 'min_effect_size': 0.0, 'mode': 'fast', 'n_used': 36, 'name': 'ADF + KPSS Joint Stationarity', 'p_adj': 0.025, 'p_raw': 0.025, 'rejected_null': True, 'stage': 'pre_model', 'statistic': -1.76392248, 'subtest': 'adf_kpss_2x2', 'target': 'Energy_kWh', 'test_id': 'T1', 'variable': 'Energy_kWh'} [artifact:annotation.assumption_fallbacks.0]; {'alpha': 0.05, 'assumptions_checked': [{'action': 'full_sample_retained', 'metric_name': 'original_n', 'metric_value': 36.0, 'name': 'large_n_subsampling_policy', 'passed': True, 'threshold': 4800}, {'action': 'canova_hansen_harmonic_f', 'metric_name': 'shapiro_p', 'metric_value': 0.02667, 'name': 'gaussian_innovations_for_harmonic_f', 'passed': True, 'threshold': 0.01}, {'action': 'switched_to_hac_newey_west', 'metric_name': 'lag1_r', 'metric_value': 0.4195, 'name': 'serial_independence', 'passed': False, 'threshold': 0.2}], 'assumptions_met': False, 'ci': {'confidence_level': 0.95, 'estimate': 0.89591324, 'high': 0.89591324, 'inference_label': 'nonparametric_exact', 'low': 0.12211574, 'method': 'percentile_bootstrap_seasonal_strength(B=120)', 'std_error': 0.11584213}, 'correction': 'holm', 'decision': 'fail', 'decision_rule': 'p_only', 'details': {'acf_2s': 0.308877, 'acf_s': 0.54827, 'autocorrelation_check': {'autocorrelated': True, 'lag1_r': 0.4195, 'ljung_box_p': 0.0002612, 'n': 36}, 'canova_hansen_f': 26.37826234, 'canova_hansen_p': 0.0, 'kruskal_wallis_h': 31.59459459, 'kruskal_wallis_p': 0.0008857, 'pacf_s': 0.112667, 'period': 12, 'policy_evaluation_note': 'p=3.813e-09 <= alpha=0.05 (p_only).', 'qs_p': 8.3e-07, 'qs_statistic': 28.01036572, 'seasonality_detected': True, 'subsampling': {'applied': False, 'max_fast_n': 4800, 'original_n': 36, 'strategy': 'full_sample_exact', 'used_n': 36}}, 'df': [4.0, 30.0], 'downstream_action': 'add_seasonal_harmonics_or_dummies', 'effect_size': 0.89591324, 'effect_size_name': 'stl_seasonal_strength_fs', 'fallback_used': 'hac_newey_west', 'family': 'stationarity_trend', 'inference_label': 'hac_robust', 'interpretation': 'Period s=12: significant unmodelled seasonality detected (canova_hansen_harmonic_f=26.378, p_raw=1.906e-09, STL Fs=0.896, ACF(s)=0.548, QS_p=8.272e-07).', 'library_versions': {'energybaseline': '3.0.0', 'numpy': '2.4.6', 'scipy': '1.17.1', 'statsmodels': '0.15.0', 'sklearn': '1.9.1', 'pandas': '3.1.1'}, 'min_effect_size': 0.0, 'mode': 'fast', 'n_used': 36, 'name': 'Seasonality Detection (ACF/PACF, Canova-Hansen & STL Strength)', 'p_adj': 3.8e-09, 'p_raw': 1.9e-09, 'rejected_null': True, 'stage': 'pre_model', 'statistic': 26.37826234, 'subtest': 'canova_hansen_harmonic_f', 'target': 'Energy_kWh', 'test_id': 'T2', 'variable': 'Energy_kWh'} [artifact:annotation.assumption_fallbacks.1]; {'alpha': 0.05, 'assumptions_checked': [{'action': 'full_sample_retained', 'metric_name': 'original_n', 'metric_value': 36.0, 'name': 'large_n_subsampling_policy', 'passed': True, 'threshold': 5000}, {'action': 'standard_ljung_box', 'metric_name': 'excess_kurtosis', 'metric_value': -0.49913, 'name': 'finite_fourth_moment_homoscedastic_innovations', 'passed': True, 'threshold': 8.0}, {'action': 'ci_via_hac_robust', 'metric_name': 'lag1_r', 'metric_value': 0.338328, 'name': 'autocorrelation_ci_calibration', 'passed': True, 'threshold': 0.2}], 'assumptions_met': True, 'ci': {'confidence_level': 0.95, 'estimate': 0.33832815, 'high': 0.6288659, 'inference_label': 'hac_robust', 'low': -0.03511675, 'method': 'bartlett_effective_n_fisher_z(n_eff=28.6)', 'std_error': 0.16938644}, 'correction': 'holm', 'decision': 'fail', 'decision_rule': 'p_only', 'details': {'durbin_watson': 1.295095, 'excess_kurtosis': -0.49913, 'lag1_r': 0.338328, 'lags_tested': 10, 'ljung_box_p': 0.0, 'ljung_box_q': 72.47812779, 'max_bartlett_band': 0.326667, 'policy_evaluation_note': 'p=1.47e-11 <= alpha=0.05 (p_only).', 'subsampling': {'applied': False, 'max_fast_n': 5000, 'original_n': 36, 'strategy': 'full_sample_exact', 'used_n': 36}}, 'df': 10.0, 'downstream_action': 'apply_hac_or_add_autoregressive_lags', 'effect_size': 0.33832815, 'effect_size_name': 'lag1_autocorrelation_r', 'fallback_used': '', 'family': 'dependence_collinearity', 'inference_label': 'hac_robust', 'interpretation': 'Ljung-Box Q(10)=72.478 (p_raw=1.47e-11), Durbin-Watson=1.295, lag-1 r=+0.338: serial autocorrelation present.', 'library_versions': {'energybaseline': '3.0.0', 'numpy': '2.4.6', 'scipy': '1.17.1', 'statsmodels': '0.15.0', 'sklearn': '1.9.1', 'pandas': '3.1.1'}, 'min_effect_size': 0.0, 'mode': 'fast', 'n_used': 36, 'name': 'Ljung-Box & Durbin-Watson Autocorrelation', 'p_adj': 0.0, 'p_raw': 0.0, 'rejected_null': True, 'stage': 'pre_model', 'statistic': 72.47812779, 'subtest': 'ljung_box_dw', 'target': 'Energy_kWh', 'test_id': 'T3', 'variable': 'Energy_kWh_residuals'} [artifact:annotation.assumption_fallbacks.2]; 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'percentile_bootstrap_nonnormality_index(B=150)', 'std_error': 0.19103462}, 'correction': 'holm', 'decision': 'pass', 'decision_rule': 'p_only', 'details': {'anderson_darling_a2': 0.382172, 'autocorrelation_check': {'autocorrelated': True, 'lag1_r': 0.338328, 'ljung_box_p': 1e-08, 'n': 36}, 'dagostino_k2': 0.510369, 'dagostino_p': 0.77477352, 'excess_kurtosis': -0.407987, 'jarque_bera_p': 0.72501536, 'jarque_bera_stat': 0.643125, 'policy_evaluation_note': 'p=1 > alpha=0.05 (p_only).', 'skewness': -0.216765, 'subsampling': {'applied': False, 'max_fast_n': 5000, 'original_n': 36, 'strategy': 'full_sample_exact', 'used_n': 36}}, 'df': 35.0, 'downstream_action': 'proceed_gaussian_parametric_intervals', 'effect_size': 0.29765823, 'effect_size_name': 'rms_skew_kurtosis_departure', 'fallback_used': 'ar1_prewhitened_innovations', 'family': 'residual_diagnostics', 'inference_label': 'hac_robust', 'interpretation': 'Normality (shapiro_wilk=0.9734, p_raw=0.5422, JB=0.643, A2=0.382, skew=-0.22, excess_kurt=-0.41): consistent with Gaussianity.', 'library_versions': {'energybaseline': '3.0.0', 'numpy': '2.4.6', 'scipy': '1.17.1', 'statsmodels': '0.15.0', 'sklearn': '1.9.1', 'pandas': '3.1.1'}, 'min_effect_size': 0.0, 'mode': 'fast', 'n_used': 36, 'name': 'Shapiro-Wilk, Jarque-Bera & Anderson-Darling Normality', 'p_adj': 1.0, 'p_raw': 0.5421638339, 'rejected_null': False, 'stage': 'pre_model', 'statistic': 0.97336859, 'subtest': 'shapiro_wilk', 'target': 'Energy_kWh', 'test_id': 'T4', 'variable': 'Energy_kWh_residuals'} [artifact:annotation.assumption_fallbacks.3]; 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{'alpha': 0.05, 'assumptions_checked': [{'action': 'halt_test_structured', 'metric_name': 'n_valid', 'metric_value': 36.0, 'name': 'minimum_sample_and_non_degeneracy', 'passed': False, 'threshold': 54}], 'assumptions_met': False, 'ci': {'confidence_level': 0.95, 'high': None, 'low': None, 'method': 'insufficient_data'}, 'correction': 'none', 'decision': 'insufficient_data', 'decision_rule': 'p_only', 'details': {}, 'df': None, 'downstream_action': 'skip_insufficient_sample', 'effect_size': None, 'effect_size_name': 'standardized_effect', 'fallback_used': '', 'family': 'breaks_regimes', 'inference_label': 'standard', 'interpretation': 'T8 (Quandt-Andrews Unknown Breakpoint sup-F Test) skipped cleanly: requires >=54 non-constant observations for trimmed Quandt-Andrews grid (trim=0.15) (n=36, required>=54).', 'library_versions': {'energybaseline': '3.0.0', 'numpy': '2.4.6', 'scipy': '1.17.1', 'statsmodels': '0.15.0', 'sklearn': '1.9.1', 'pandas': '3.1.1'}, 'min_effect_size': 0.0, 'mode': 'fast', 'n_used': 36, 'name': 'Quandt-Andrews Unknown Breakpoint sup-F Test', 'p_adj': None, 'p_raw': None, 'rejected_null': False, 'stage': 'pre_model', 'statistic': None, 'subtest': 'precheck_halted', 'target': 'Energy_kWh', 'test_id': 'T8', 'variable': 'Energy_kWh'} [artifact:annotation.assumption_fallbacks.7]; {'alpha': 0.05, 'assumptions_checked': [{'action': 'full_sample_retained', 'metric_name': 'original_n', 'metric_value': 36.0, 'name': 'large_n_subsampling_policy', 'passed': True, 'threshold': 5000}, {'action': 'pettitt_asymptotic_rank', 'metric_name': 'tie_fraction', 'metric_value': 0.0, 'name': 'continuous_distribution_low_ties', 'passed': True, 'threshold': 0.35}, {'action': 'switched_to_moving_block_bootstrap', 'metric_name': 'lag1_r', 'metric_value': 0.425578, 'name': 'serial_independence', 'passed': False, 'threshold': 0.2}], 'assumptions_met': False, 'ci': {'confidence_level': 0.95, 'estimate': -0.32698413, 'high': 0.70897569, 'inference_label': 'block_bootstrap', 'low': -0.62222222, 'method': 'moving_block_bootstrap_percentile(b=3,B=150)', 'std_error': 0.46675636}, 'correction': 'holm', 'decision': 'pass', 'decision_rule': 'p_only', 'details': {'asymptotic_p': 0.5303037773, 'autocorrelation_check': {'autocorrelated': True, 'lag1_r': 0.425578, 'ljung_box_p': 0.0014177, 'n': 36}, 'change_point_1based': 15, 'median_after': 16083.0, 'median_before': 19072.0, 'normalized_median_shift': -0.760918, 'policy_evaluation_note': 'p=1 > alpha=0.05 (p_only).', 'subsampling': {'applied': False, 'max_fast_n': 5000, 'original_n': 36, 'strategy': 'full_sample_exact', 'used_n': 36}, 'tie_fraction': 0.0}, 'df': 35.0, 'downstream_action': 'retain_homogeneous_series', 'effect_size': -0.32698413, 'effect_size_name': 'pettitt_rank_biserial_r', 'fallback_used': 'moving_block_bootstrap_pettitt', 'family': 'breaks_regimes', 'inference_label': 'block_bootstrap', 'interpretation': 'Pettitt K_T=103.00 at tau=15 (p_raw=0.61, asymptotic_p=0.5303, median 1.91e+04->1.61e+04, rank_biserial=-0.327): no significant level shift.', 'library_versions':

{'energybaseline': '3.0.0', 'numpy': '2.4.6', 'scipy': '1.17.1', 'statsmodels': '0.15.0', 'sklearn': '1.9.1', 'pandas': '3.1.1'}, 'min_effect_size': 0.0, 'mode': 'fast', 'n_used': 36, 'name': 'Pettitt Non-Parametric Change-Point Test', 'p_adj': 1.0, 'p_raw': 0.61, 'rejected_null': False, 'stage': 'pre_model', 'statistic': 103.0, 'subtest': 'pettitt_bootstrap_fallback', 'target': 'Energy_kWh', 'test_id': 'T9', 'variable': 'Energy_kWh'} [artifact:annotation.assumption_fallbacks.8]; {'alpha': 0.05, 'assumptions_checked': [{'action': 'full_sample_retained', 'metric_name': 'original_n', 'metric_value': 18.0, 'name': 'large_n_subsampling_policy', 'passed': True, 'threshold': 5000}, {'action': 'dm_hln_parametric', 'metric_name': 'excess_kurtosis', 'metric_value': 1.914031, 'name': 'loss_differential_finite_variance', 'passed': True, 'threshold': 12.0}, {'action': 'standard_inference', 'metric_name': 'lag1_r', 'metric_value': 0.253209, 'name': 'serial_independence', 'passed': True, 'threshold': 0.2}], 'assumptions_met': True, 'ci': {'confidence_level': 0.95, 'estimate': -0.16106114, 'high': 0.32221622, 'inference_label': 'standard', 'low': -0.6443385, 'method': 'hln_bartlett_cohen_d_t', 'std_error': 0.22906142}, 'correction': 'holm', 'decision': 'pass', 'decision_rule': 'p_only', 'details': {'autocorrelation_check': {'autocorrelated': False, 'lag1_r': 0.253209, 'ljung_box_p': 0.18072777, 'n': 18}, 'dm_raw_statistic': -0.70313516, 'h': 1, 'hln_statistic': -0.68332455, 'hln_p': 0.503609314, 'mean_loss_diff': -29075.14193741, 'policy_evaluation_note': 'p=1 > alpha=0.05 (p_only).', 'power': 2, 'subsampling': {'applied': False, 'max_fast_n': 5000, 'original_n': 18, 'strategy': 'full_sample_exact', 'used_n': 18}, 'mcs': {'alpha': 0.05, 'eliminated_models': ['Degree-day regression', 'Lasso regression', 'Bayesian ridge', 'Ridge regression'], 'elimination_sequence': ['Degree-day regression', 'Lasso regression', 'Bayesian ridge', 'Ridge regression'], 'mcs_pvalues': {'Degree-day regression': 0.0, 'Lasso regression': 0.0, 'Bayesian ridge': 0.0, 'Ridge regression': 0.0, 'Multiple linear regression (MLR)': 1.0}, 'mean_losses': {'Degree-day regression': None, 'Lasso regression': None, 'Bayesian ridge': None, 'Ridge regression': None, 'Multiple linear regression (MLR)': None}, 'superior_set': ['Multiple linear regression (MLR)'], 't_max_initial': None}}, 'df': 17.0, 'downstream_action': 'retain_equivalent_forecast_models', 'effect_size': -0.16106114, 'effect_size_name': 'loss_diff_cohen_d', 'fallback_used': '', 'family': 'model_comparison', 'inference_label': 'standard', 'interpretation': 'Diebold-Mariano (dm_hln=-0.6833, p_raw=0.5036, mean_loss_diff=-2.908e+04, Cohen_d=-0.161): models are statistically indistinguishable on OOF loss.', 'library_versions': {'energybaseline': '3.0.0', 'numpy': '2.4.6', 'scipy': '1.17.1', 'statsmodels': '0.15.0', 'sklearn': '1.9.1', 'pandas': '3.1.1'}, 'min_effect_size': 0.0, 'mode': 'fast', 'n_used': 18, 'name': 'Diebold-Mariano (HLN) OOF Forecast Equivalence & MCS', 'p_adj': 1.0, 'p_raw': 0.503609314, 'rejected_null': False, 'stage': 'post_model', 'statistic': -0.68332455, 'subtest': 'dm_hln', 'target': 'Energy_kWh', 'test_id': 'T10', 'variable': 'Degree-day regression_vs_Lasso regression'} [artifact:annotation.assumption_fallbacks.9]; 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{'alpha': 0.05, 'assumptions_checked': [{'action': 'full_sample_retained', 'metric_name': 'original_n', 'metric_value': 36.0, 'name': 'large_n_subsampling_policy', 'passed': True, 'threshold': 4000}], 'assumptions_met': True, 'ci': {'confidence_level': 0.95, 'estimate': 0.0, 'high': 0.0, 'inference_label': 'standard', 'low': 0.0, 'method': 'complete_data_exact', 'std_error': 0.0}, 'correction': 'holm', 'decision': 'pass', 'decision_rule': 'p_only', 'details': {'missing_cells': 0, 'subsampling': {'applied': False, 'max_fast_n': 4000, 'original_n': 36, 'strategy': 'full_sample_exact', 'used_n': 36}, 'policy_evaluation_note': 'p=1 > alpha=0.05 (p_only).'}, 'df': 0.0, 'downstream_action': 'proceed_complete_case', 'effect_size': 0.0, 'effect_size_name': 'little_mcar_cramers_phi', 'fallback_used': '', 'family': 'missingness_balance', 'inference_label': 'standard', 'interpretation': 'Dataset has 0 missing numeric cells; MCAR holds trivially (d^2=0, p_raw=1.0).', 'library_versions': {'energybaseline': '3.0.0', 'numpy': '2.4.6', 'scipy': '1.17.1', 'statsmodels': '0.15.0', 'sklearn': '1.9.1', 'pandas': '3.1.1'}, 'min_effect_size': 0.0, 'mode': 'fast', 'n_used': 36, 'name': "Little's Multivariate MCAR Missingness Test", 'p_adj': 1.0, 'p_raw': 1.0, 'rejected_null': False, 'stage': 'post_model', 'statistic': 0.0, 'subtest': 'complete_data_no_missingness', 'target': 'Energy_kWh', 'test_id': 'T12', 'variable': 'Energy_kWh,Avg_Temp_C'} [artifact:annotation.assumption_fallbacks.11]; {'alpha': 0.05, 'assumptions_checked': [{'action': 'full_sample_retained', 'metric_name': 'original_n', 'metric_value': 36.0, 'name': 'large_n_subsampling_policy', 'passed': True, 'threshold': 6000}, {'action': 'bonferroni_studentized_cooks_d', 'metric_name': 'trimmed_90pct_shapiro_p', 'metric_value': 0.174515, 'name': 'core_bulk_gaussianity_for_studentized_bonferroni', 'passed': True, 'threshold': 0.01}, {'action': 'flag_only_preserve_all_rows', 'metric_name': 'rows_deleted', 'metric_value': 0.0, 'name': 'flag_only_zero_mutation_policy', 'passed': True, 'threshold': 0}, {'action': 'switched_to_hac_newey_west', 'metric_name': 'lag1_r', 'metric_value': 0.338328, 'name': 'serial_independence', 'passed': False, 'threshold': 0.2}], 'assumptions_met': False, 'ci': {'confidence_level': 0.95, 'estimate': 0.22002687, 'high': 1.12789802, 'inference_label': 'nonparametric_exact', 'low': 0.093157, 'method': 'percentile_bootstrap_max_cooks_d(B=100)', 'std_error': 0.29800824}, 'correction': 'holm', 'decision': 'pass', 'decision_rule': 'p_only', 'details': {'autocorrelation_check': {'autocorrelated': True, 'lag1_r': 0.338328, 'ljung_box_p': 1e-08, 'n': 36}, 'bonferroni_outlier_p': 1.0, 'cooks_cutoff_4_over_n': 0.11111111, 'flag_only': True, 'flagged_count': 0, 'flagged_indices': [], 'leverage_max': 0.09447474, 'max_abs_ext_studentized': 2.23866186, 'max_cooks_d': 0.22002687, 'max_cooks_index': 6, 'max_robust_z': 2.13157985,

'policy_evaluation_note': 'p=1 > alpha=0.05 (p_only).', 'rows_deleted': 0, 'subsampling': {'applied': False, 'max_fast_n': 6000, 'original_n': 36, 'strategy': 'full_sample_exact', 'used_n': 36}}, 'df': 33.0, 'downstream_action': 'no_influential_outliers_flagged', 'effect_size': 0.22002687, 'effect_size_name': 'max_cooks_distance', 'fallback_used': 'hac_newey_west', 'family': 'residual_diagnostics', 'inference_label': 'hac_robust', 'interpretation': "Flag-only influence check: max Cook's D=0.2200 (4/n=0.1111), max |studentized t|=2.239 (Bonferroni p_raw=1), flagged_rows=0 (0 rows deleted).", 'library_versions': {'energybaseline': '3.0.0', 'numpy': '2.4.6', 'scipy': '1.17.1', 'statsmodels': '0.15.0', 'sklearn': '1.9.1', 'pandas': '3.1.1'}, 'min_effect_size': 0.0, 'mode': 'fast', 'n_used': 36, 'name': "Flag-Only Outlier & Cook's Distance Influence Diagnostics", 'p_adj': 1.0, 'p_raw': 1.0, 'rejected_null': False, 'stage': 'pre_model', 'statistic': 2.23866186, 'subtest': 'cooks_distance_bonferroni_flag_only', 'target': 'Energy_kWh', 'test_id': 'T13', 'variable': 'Energy_kWh'} [artifact:annotation.assumption_fallbacks.12]; {'alpha': 0.05, 'assumptions_checked': [{'action': 'not applicable (no group column)', 'metric_name': 'group_column', 'metric_value': 'none', 'name': 'declared_group_column', 'passed': False, 'threshold': 'present'}], 'assumptions_met': False, 'ci': {'confidence_level': 0.95, 'high': None, 'low': None, 'method': 'not_applicable'}, 'correction': 'none', 'decision': 'not applicable (no group column)', 'decision_rule': 'p_only', 'details': {}, 'df': None, 'downstream_action': 'skip_causal_group_mv', 'effect_size': None, 'effect_size_name': 'standardized_effect', 'fallback_used': '', 'family': 'causal_group', 'inference_label': 'standard', 'interpretation': "T14 causal M&V tests require a declared 'group' column (treatment vs control); none present.", 'library_versions': {'energybaseline': '3.0.0', 'numpy': '2.4.6', 'scipy': '1.17.1', 'statsmodels': '0.15.0', 'sklearn': '1.9.1', 'pandas': '3.1.1'}, 'min_effect_size': 0.0, 'mode': 'fast', 'n_used': 0, 'name': 'Causal M&V Group Balance & Parallel Trends', 'p_adj': None, 'p_raw': None, 'rejected_null': False, 'stage': 'causal_group', 'statistic': None, 'subtest': 'group_column_absent', 'target': 'Energy_kWh', 'test_id': 'T14', 'variable': ''} [artifact:annotation.assumption_fallbacks.13]

Key Metrics

Artifact ID	Value	Metric	Unit
[artifact:metric.t1.n_obs]	36	Energy_kWh / Observations	observations
[artifact:metric.t1.quality]	100.0	Energy_kWh / Quality score	score
[artifact:metric.t1.completeness]	1.0	Energy_kWh / Completeness	fraction
[artifact:metric.t1.validity]	1.0	Energy_kWh / Validity	fraction
[artifact:metric.t1.uniqueness]	1.0	Energy_kWh / Uniqueness	fraction
[artifact:metric.t1.cvrmse]	3.938894923882289	Energy_kWh / CV(RMSE)	%
[artifact:metric.t1.nmbe]	1.3608802161860205	Energy_kWh / NMBE	%
[artifact:metric.t1.r_squared]	0.9548356980149817	Energy_kWh / R-squared	dimensionless
[artifact:metric.t1.rmse]	653.6464829685196	Energy_kWh / RMSE	kWh
[artifact:metric.t1.mae]	564.856481488332	Energy_kWh / MAE	kWh
[artifact:metric.t1.mape]	3.6939938348498997	Energy_kWh / MAPE	%
[artifact:metric.t1.smape]	3.7047109254919044	Energy_kWh / sMAPE	%
[artifact:metric.t1.aic]	Unavailable	Energy_kWh / AIC	score
[artifact:metric.t1.bic]	Unavailable	Energy_kWh / BIC	score
[artifact:metric.t1.n_params]	3	Energy_kWh / Parameter count	parameters
[artifact:metric.t1.score]	95.94587809945104	Energy_kWh / Composite score	score
[artifact:metric.t1.n_models]	19	Energy_kWh / Evaluated models	models
[artifact:metric.t1.n_qualified]	15	Energy_kWh / Qualified models	models
[artifact:metric.t1.n_features]	5	Energy_kWh / Selected features	features
[artifact:metric.t1.actual_mean]	16943.583333333332	Energy_kWh / Observed mean	kWh

Executive Summary

Purpose: Document the evidence available for this analysis stage

Method: Persisted analysis artifacts; no statistical inference during rendering

Settings: See the linked configuration and provenance tables

Results: See artifact-linked tables and statements

Charts: Charts are omitted when their required observations are unavailable

Interpretation: Review evidence in its declared units and validity domain

Decision: Engineering review is required before changing a locked baseline

Caveats: Missing evidence is disclosed, not replaced with inferred results

Recorded metric: 36 [artifact:metric.t1.n_obs]

Recorded metric: 100.0 [artifact:metric.t1.quality]

Recorded metric: 1.0 [artifact:metric.t1.completeness]

Recorded metric: 1.0 [artifact:metric.t1.validity]

Recorded metric: 1.0 [artifact:metric.t1.uniqueness]

Recorded metric: 3.938894923882289 [artifact:metric.t1.cvrmse]

Recorded metric: 1.3608802161860205 [artifact:metric.t1.nmbe]

Recorded metric: 0.9548356980149817 [artifact:metric.t1.r_squared]

Recorded metric: 653.6464829685196 [artifact:metric.t1.rmse]

Recorded metric: 564.856481488332 [artifact:metric.t1.mae]

Recorded metric: 3.6939938348498997 [artifact:metric.t1.mape]

Recorded metric: 3.7047109254919044 [artifact:metric.t1.smape]

Recorded metric: None [artifact:metric.t1.aic]

Recorded metric: None [artifact:metric.t1.bic]

Recorded metric: 3 [artifact:metric.t1.n_params]

Recorded metric: 95.94587809945104 [artifact:metric.t1.score]

Recorded metric: 19 [artifact:metric.t1.n_models]

Recorded metric: 15 [artifact:metric.t1.n_qualified]

Recorded metric: 5 [artifact:metric.t1.n_features]

Recorded metric: 16943.583333333332 [artifact:metric.t1.actual_mean]

Selected Baseline & UQ

Purpose: Document the evidence available for this analysis stage

Method: Persisted analysis artifacts; no statistical inference during rendering

Settings: See the linked configuration and provenance tables

Results: See artifact-linked tables and statements

Charts: Charts are omitted when their required observations are unavailable

Interpretation: Review evidence in its declared units and validity domain

Decision: Engineering review is required before changing a locked baseline

Caveats: Missing evidence is disclosed, not replaced with inferred results

Evidence availability is recorded [artifact:availability.selected_baseline]

Reproducibility

Purpose: Document the evidence available for this analysis stage

Method: Persisted analysis artifacts; no statistical inference during rendering

Settings: See the linked configuration and provenance tables

Results: See artifact-linked tables and statements

Charts: Charts are omitted when their required observations are unavailable

Interpretation: Review evidence in its declared units and validity domain

Decision: Engineering review is required before changing a locked baseline

Caveats: Missing evidence is disclosed, not replaced with inferred results

Evidence availability is recorded [artifact:availability.reproducibility]

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python	3.11.2

Disclosures

Purpose: Document the evidence available for this analysis stage

Method: Persisted analysis artifacts; no statistical inference during rendering

Settings: See the linked configuration and provenance tables

Results: See artifact-linked tables and statements

Charts: Charts are omitted when their required observations are unavailable

Interpretation: Review evidence in its declared units and validity domain

Decision: Engineering review is required before changing a locked baseline

Caveats: Missing evidence is disclosed, not replaced with inferred results

Evidence availability is recorded [artifact:availability.disclosures]

detail
{'alpha': 0.05, 'assumptions_checked': [{'action': 'full_sample_retained', 'metric_name': 'original_n', 'metric_value': 36.0, 'name': 'large_n_subsampling_policy', 'passed': True, 'threshold': 4000}, {'action': 'proceed_adf_kpss', 'metric_name': 'std', 'metric_value': 3171.830633, 'name': 'non_degenerate_variance', 'passed': True, 'threshold': 1e-12}, {'action': 'switched_to_hac_newey_west', 'metric_name': 'lag1_r', 'metric_value': 0.4195, 'name': 'serial_independence', 'passed': False, 'threshold': 0.2}], 'assumptions_met': False, 'ci': {'confidence_level': 0.95, 'estimate': 0.426212, 'high': 0.69053419, 'inference_label': 'hac_robust', 'low': 0.16188982, 'method': 'hac_ar1_persistence_t(bw=3)', 'std_error': 0.1223505}, 'correction': 'holm', 'decision': 'fail', 'decision_rule': 'p_only', 'details': {'adf_lags': 10, 'adf_p': 0.39851449, 'adf_statistic': -1.76392248, 'autocorrelation_check': {'autocorrelated': True, 'lag1_r': 0.4195, 'ljung_box_p': 0.0002612, 'n': 36}, 'kpss_lags': 1, 'kpss_p': 0.1, 'kpss_statistic': 0.11648566, 'lock_custom_matrix_decision': True, 'matrix_cell':

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'Heteroskedasticity (white_quadratic_lm=9.915, p_raw=0.00703, BP=6.497, White=9.915, ARCH(3)=1.673, sqrt(R2_aux)=0.507): non-constant error variance detected.', 'library_versions': {'energybaseline': '3.0.0', 'numpy': '2.4.6', 'scipy': '1.17.1', 'statsmodels': '0.15.0', 'sklearn': '1.9.1', 'pandas': '3.1.1'}, 'min_effect_size': 0.0, 'mode': 'fast', 'n_used': 36, 'name': 'Breusch-Pagan, White & ARCH-LM Heteroskedasticity', 'p_adj': 0.0210892544, 'p_raw': 0.0070297515, 'rejected_null': True, 'stage': 'pre_model', 'statistic': 9.91520785, 'subtest': 'white_quadratic_lm', 'target': 'Energy_kWh', 'test_id': 'T5', 'variable': 'Energy_kWh_residuals'}